

Scanned with CamScanner

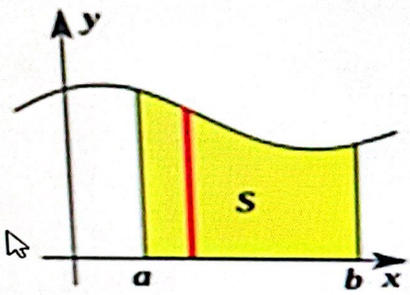
In this section, we will learn about:

Using integration to find out the volume of a solid.

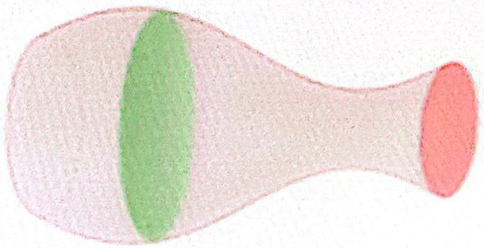


Scanned with CamScanner

Integration



1D → 2D



2D → 3D

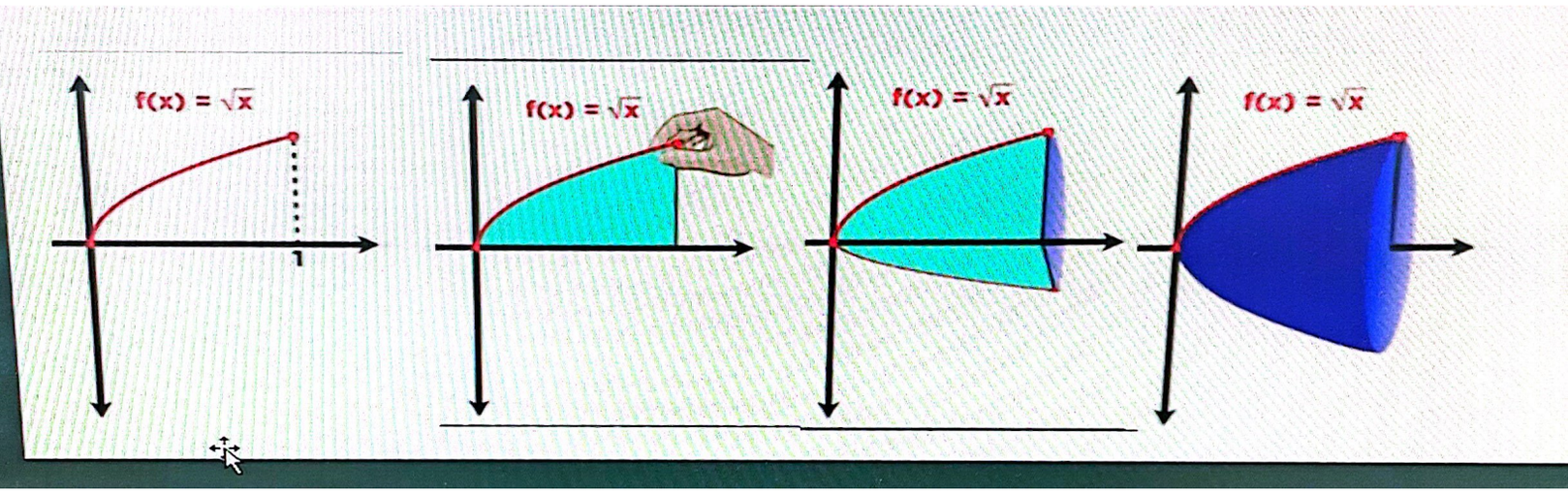
to add notes
Notes
Comments
73%
37°C مشمس
2:53 PM
9/27/2022
Scanned with CamScanner



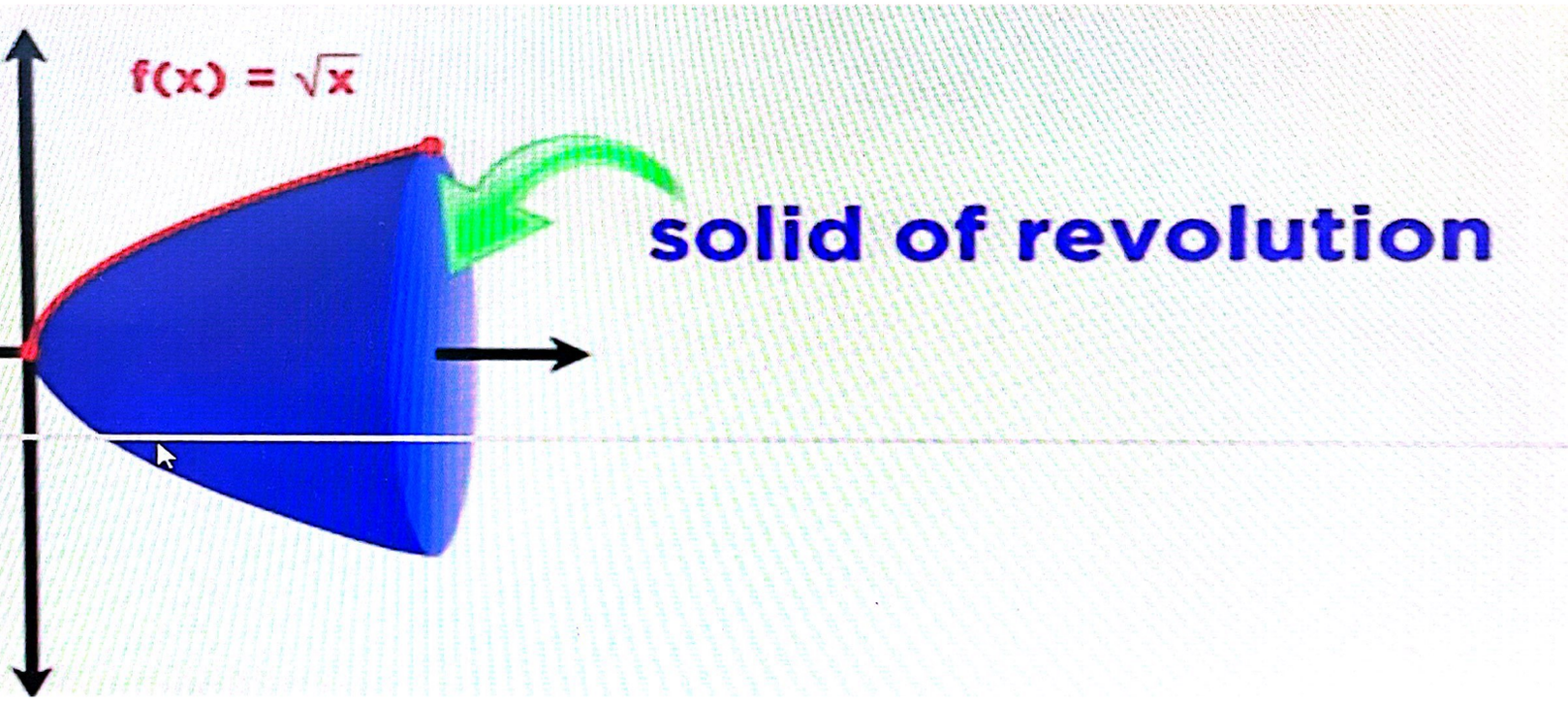
Example

Find the volume of the solid obtained by
rotating about the x-axis the region under
the curve $y=\sqrt{x}$ from 0 to 4.

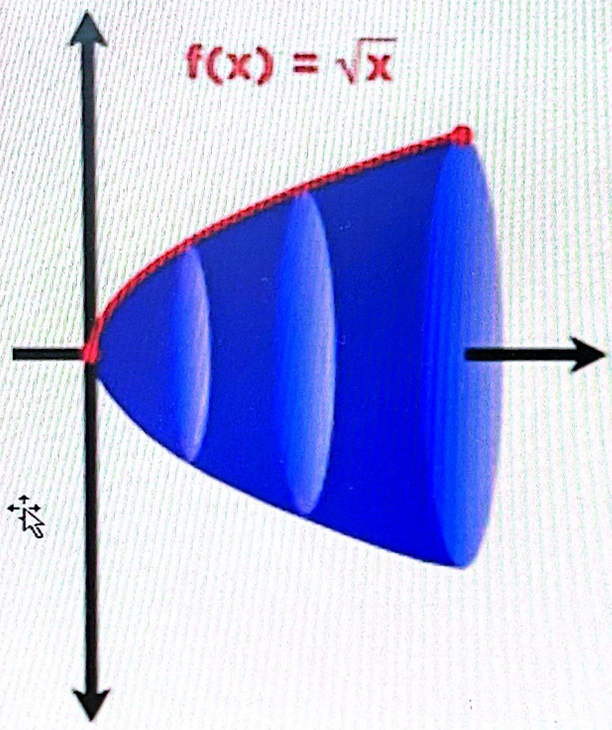
Scanned with CamScanner



Scanned with CamScanner



Scanned with CamScanner



let's add up all the
cross sections

$$\int_0^1 A(x)dx$$

But

$$A(x)=\pi R^2(x)$$

Scanned with CamScanner



Remark:

*

$$V = \int_a^b A(x)\,dx \quad \text{or} \quad V = \int_c^d A(y)\,dy$$

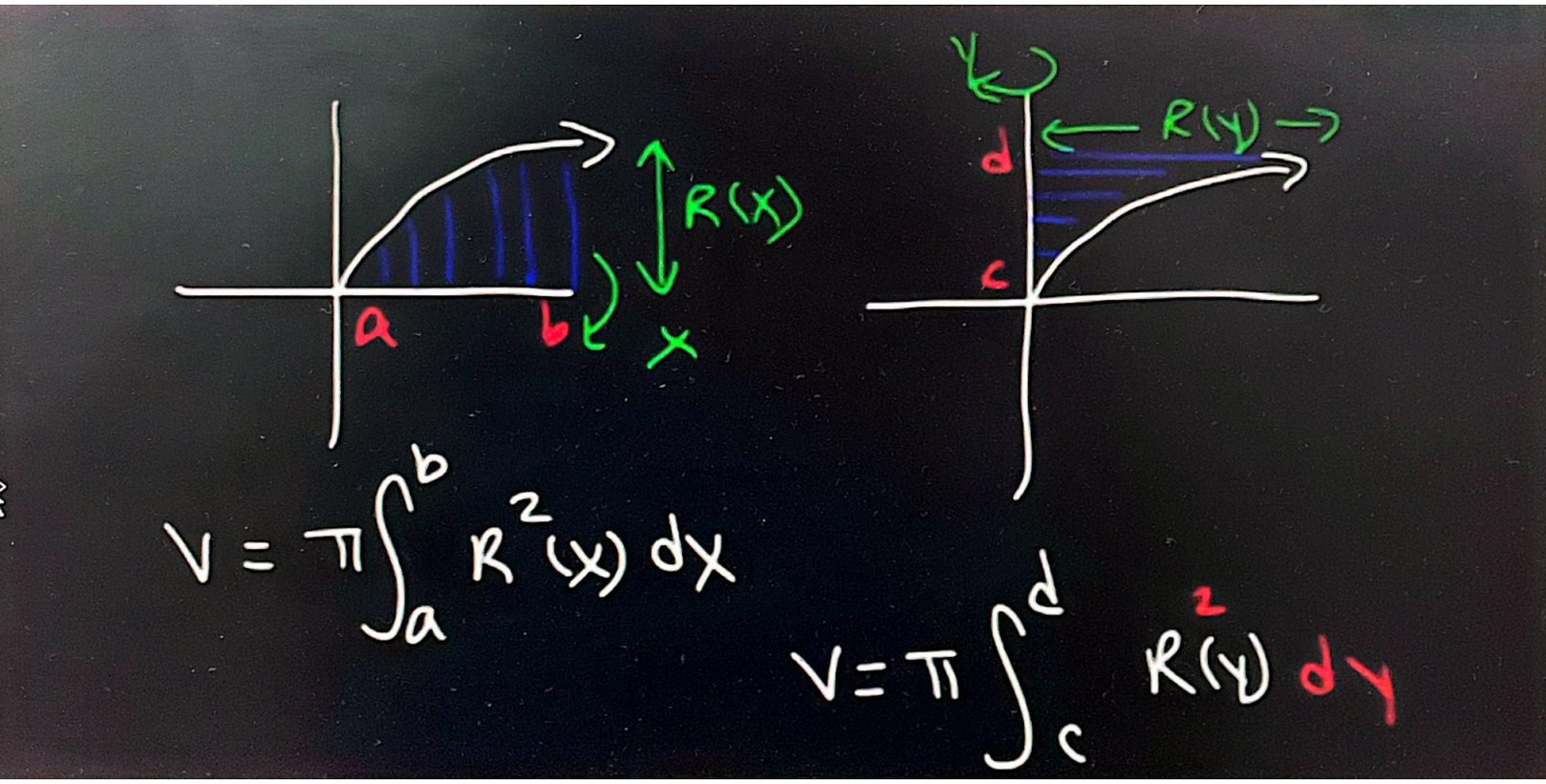
where $A = \pi(R)^2$

***R is the distance between the curve and the axis of revolution.**

***If the axis of revolution is x- axis or parallel to it then R should be in term of x.**

***If the axis of revolution is y- axis or parallel to it then R should be in term of y.**

Scanned with CamScanner

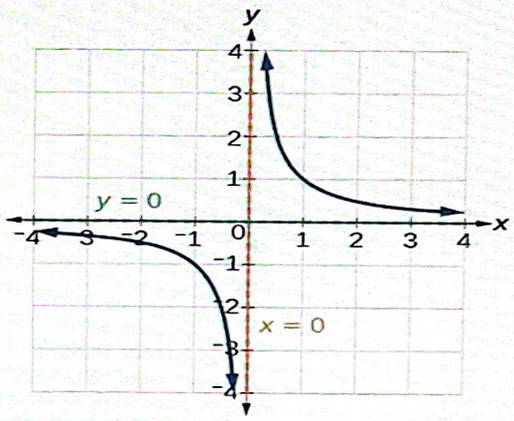


Scanned with CamScanner



Example:

Find the volume of the solid obtained
by rotating the region bounded by
$= 1/x$,
$x=1$, and $x = 3$ about the x-axis.



Scanned with CamScanner

Find the volume of the solid obtained

by rotating the region bounded by $y = x^2$,

$Y = 4$, and $x = 0$ around the y-axis.



Example:

Scanned with CamScanner

6.2 volumes - PowerPoint

Transitions Animations Slide Show Review View Tell me what you want to do... Sign in Share

Font Paragraph Drawing Editing

Arrange Quick Styles Shape Fill Shape Outline Shape Effects Find Replace Select



Example:*

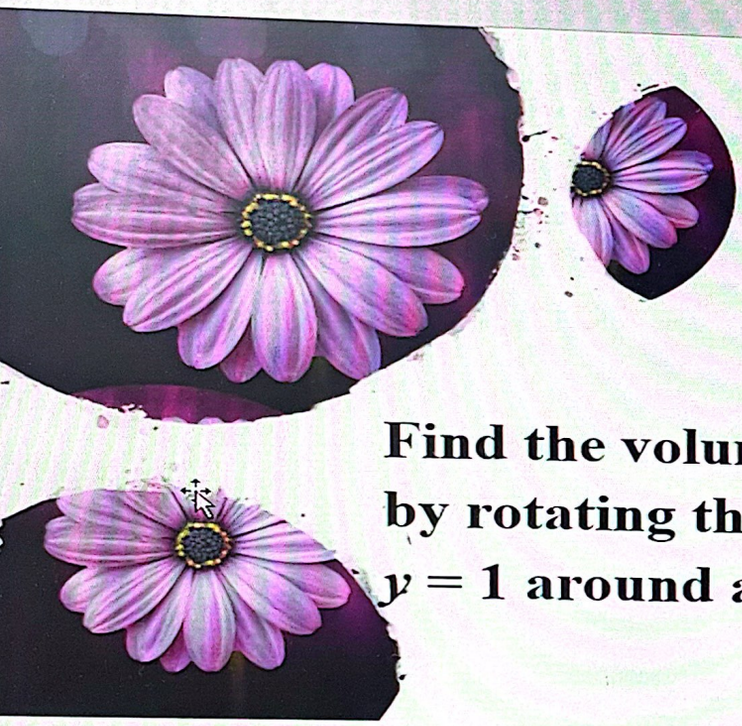
Find the volume of the solid obtained by rotating the region bounded by $y = x^2$, $Y = 4$, and $x = 0$ around the $y=4$.

e idea like next example

Notes Comments 73%

37°C مشمس 2:53 PM 9/27/2022

Scanned with CamScanner



Example*:

Find the volume of the solid obtained by rotating the region bounded by $y = 2-x^2$, $y = 1$ around about y=1.

to add notes

Notes Comments 73%

37°C مشمس 2:53 PM 9/27/2022

Scanned with CamScanner

Font Paragraph Drawing Editing
Arrange Quick Styles Shape Fill Shape Outline Shape Effects Find Replace Select

Example:





The region R enclosed by the curve and $y = \sqrt{x}$, y=0 and x=3 is rotated about the x=3.

Find the volume of the resulting solid

k to add notes
Notes Comments 73%
Scanned with CamScanner

6.2 volumes - PowerPoint

Transitions Animations Slide Show Review View Tell me what you want to do... Sign in Share

Font Paragraph Drawing Editing

Arrange Quick Styles Shape Fill Shape Outline Shape Effects Find Replace Select



Example:

Find the volume of the solid obtained by rotating the region bounded by $y = x^3$, $Y = 8$, and $x = 0$ about the y-axis.

to add notes

Notes Comments

Scanned with CamScanner

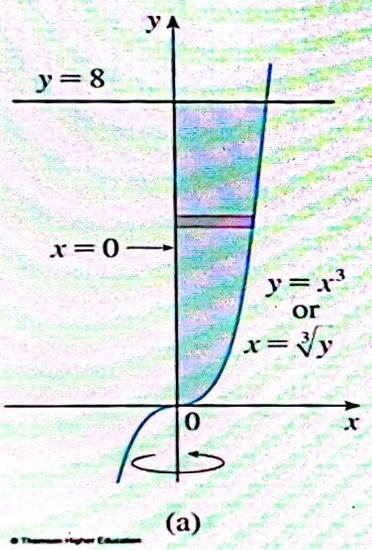
6.2 volumes - PowerPoint

Transitions Animations Slide Show Review View Tell me what you want to do... Sign in Share

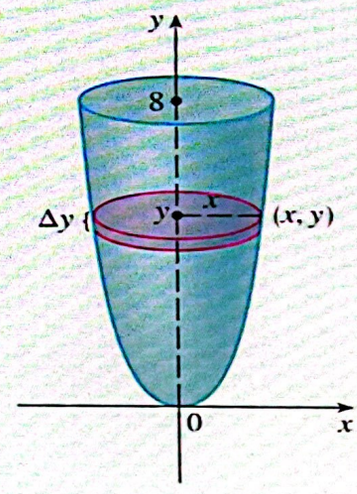
Font Paragraph Drawing Editing

Shape Fill Shape Outline Shape Effects Arrange Quick Styles Find Replace Select

$y = x^3$



(a)

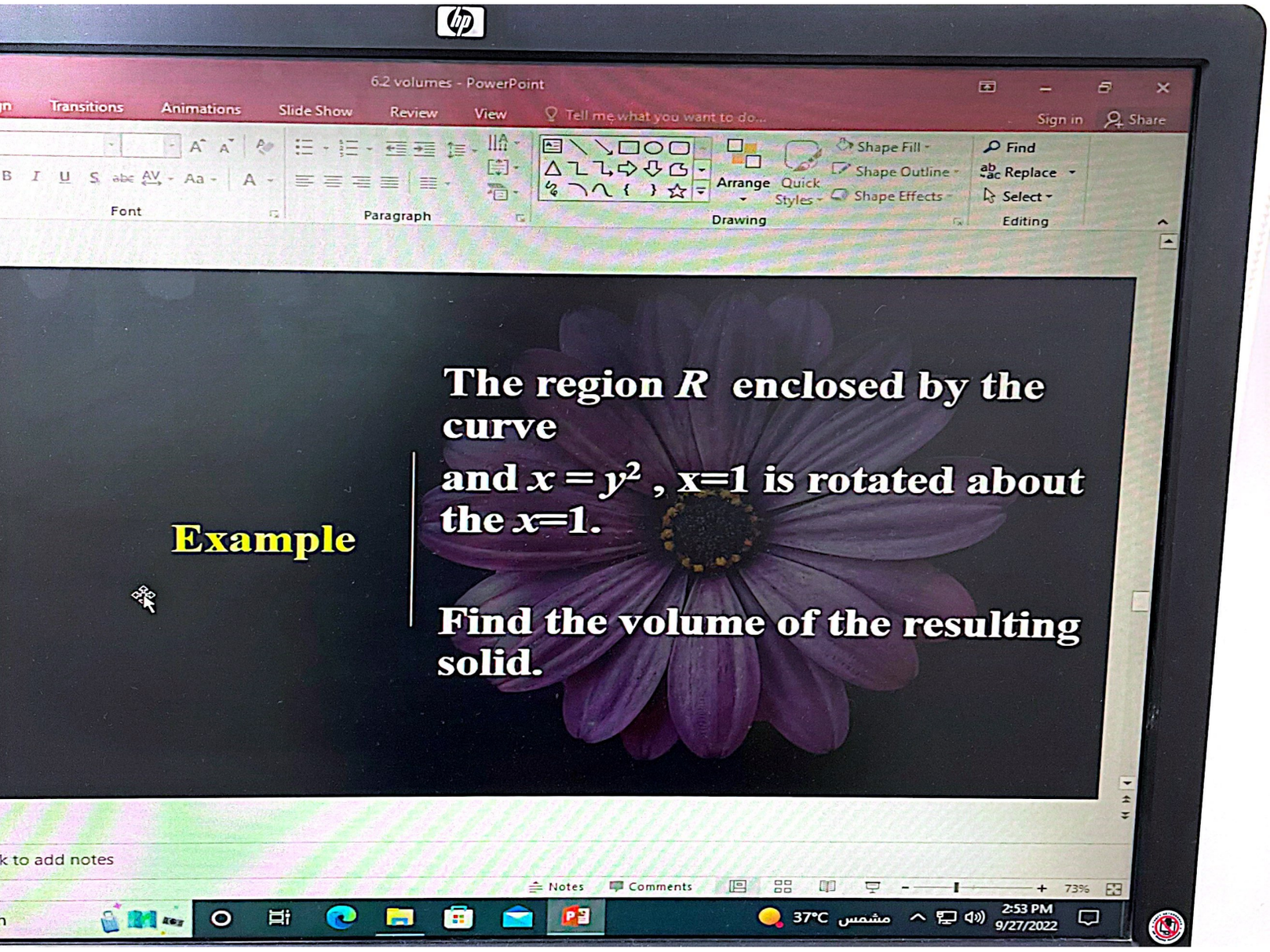


(b)

to add notes

Notes Comments 73%

Scanned with CamScanner



Scanned with CamScanner

6.2 volumes - PowerPoint

Transitions Animations Slide Show Review View Tell me what you want to do... Sign in Share

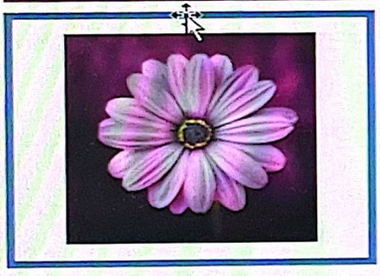
Font Paragraph Drawing Editing

Shape Fill Shape Outline Shape Effects Arrange Quick Styles Find Replace Select

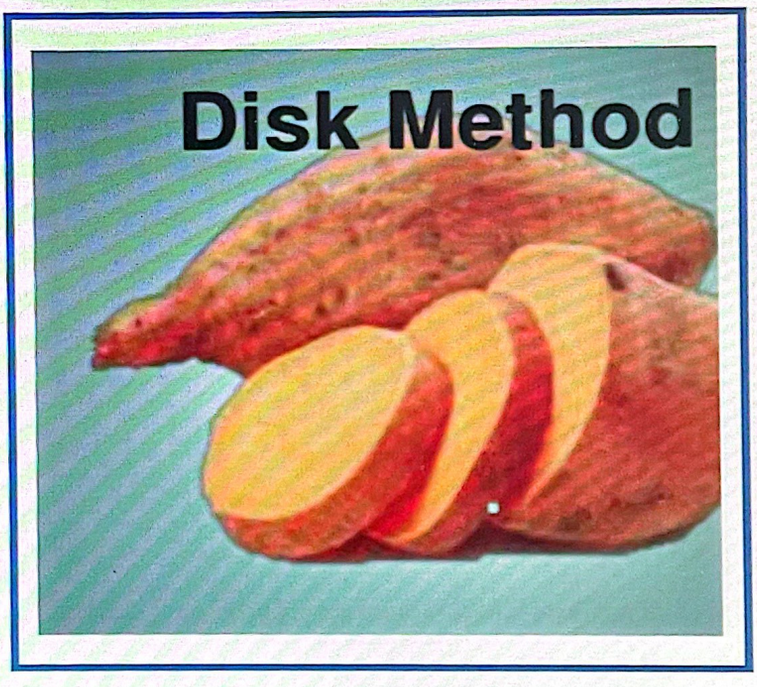
Divide the region into n equal disks

Find the volume of each disk, to get an approximate volume.

As the number of disks approaches infinity, we approach the actual volume.







Scanned with CamScanner

The method to find volume of solid generated by revolving the region bounded between two curves is known washer method



to add notes

Notes Comments 73%

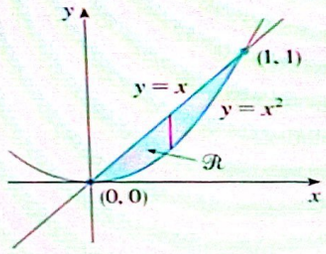
37°C مشمس 2:53 PM 9/27/2022

Scanned with CamScanner

Example 1

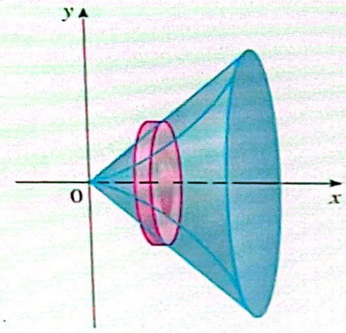
The region R enclosed by the curves $y = x$ and $y = x^2$ is rotated about the x-axis.

Find the volume of the resulting solid.

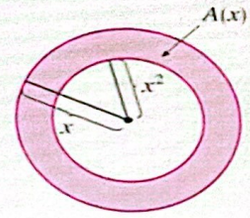


(a)

© Thomson Higher Education



(b)

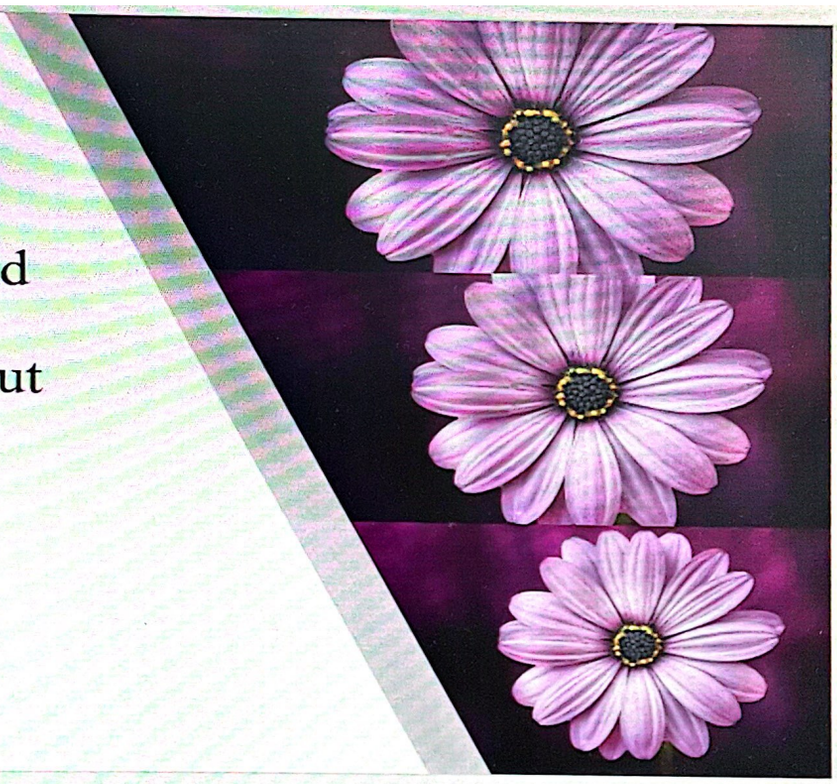


(c)

Scanned with CamScanner

Example 2:

Find the volume of the solid obtained by rotating the region in last Example about the line $y = 2$.



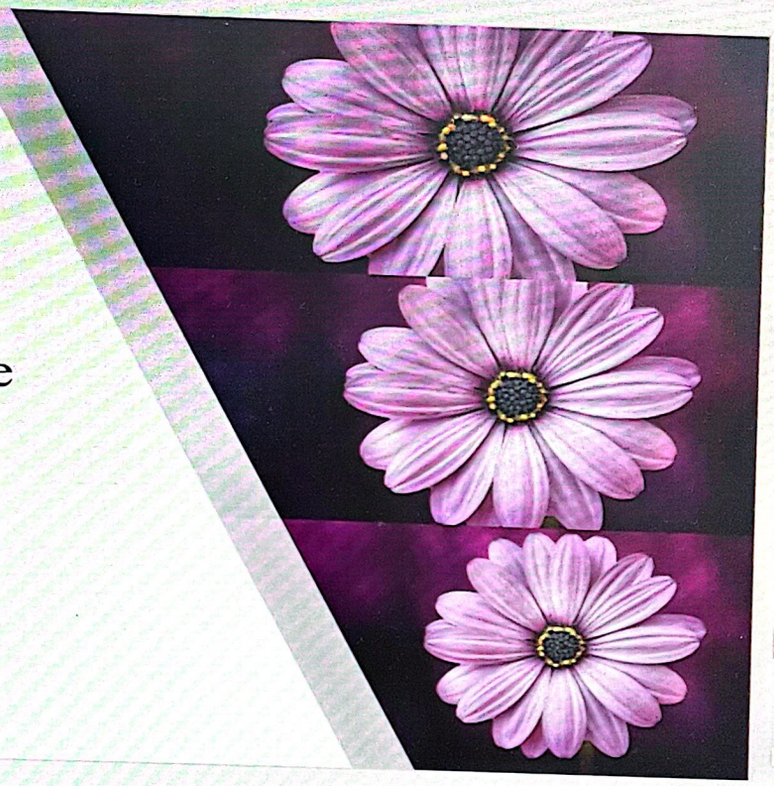
to add notes

Notes Comments 73%

Scanned with CamScanner

Example 3:

Find the volume of the solid obtained by rotating the region in Example 1 about the line x=-1.



ck to add notes

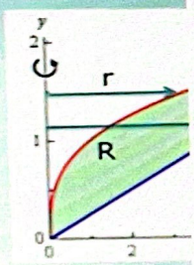
Notes Comments 73%

37°C مشمس 2:54 PM 9/27/2022

Scanned with CamScanner

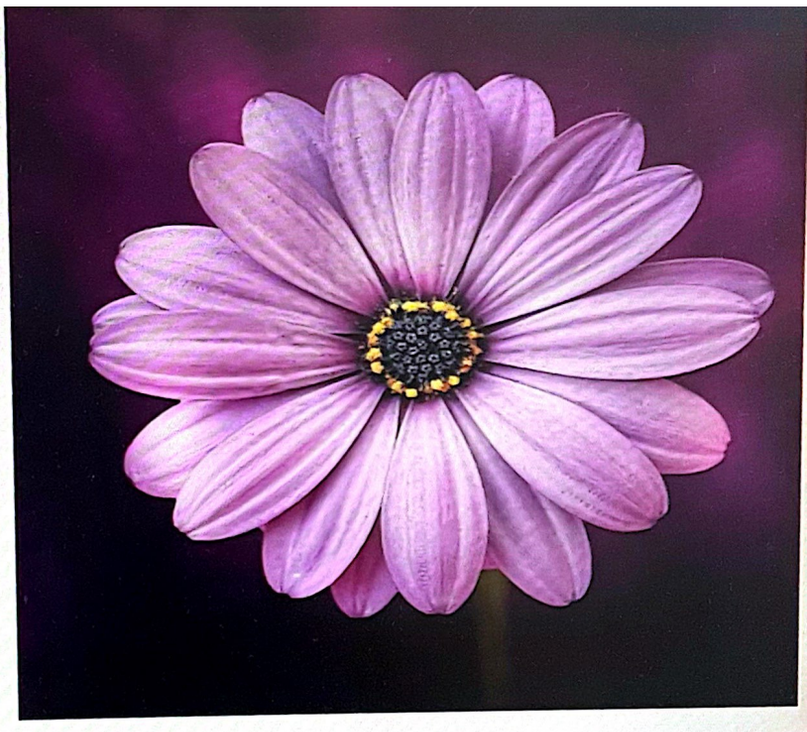
Setting up the Equatio

$$A = \pi\left(\left(\begin{matrix}\text{outer}\\ \text{radius}\end{matrix}\right)^2 - \left(\begin{matrix}\text{inner}\\ \text{radius}\end{matrix}\right)^2\right)$$

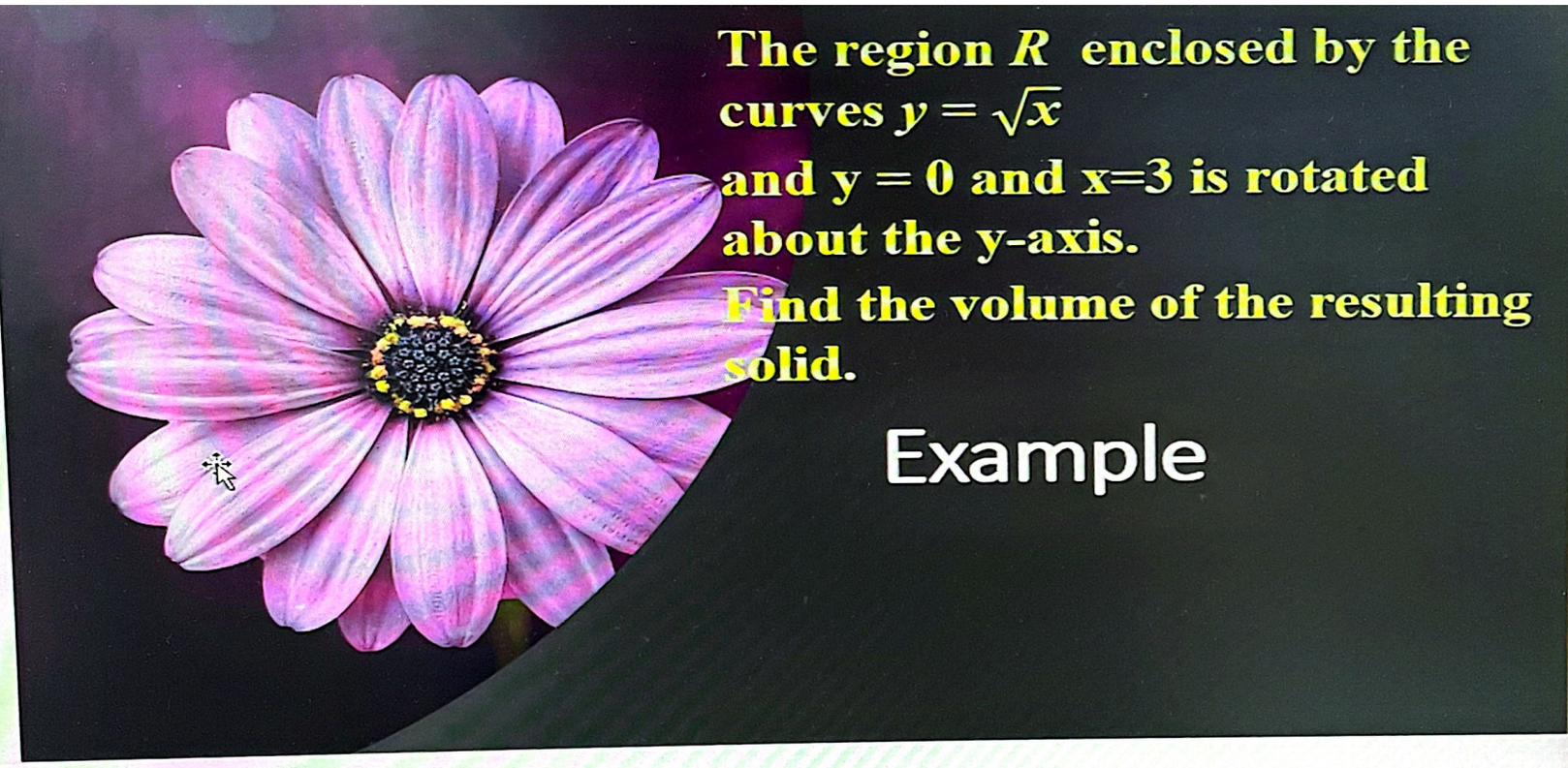


Outer Function

Inner Function



Scanned with CamScanner



Scanned with CamScanner